

3-Month (90-Day) DevOps Study + Interview Preparation Plan

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Duration: 3 Months (90 Days)

Daily Study Time: 2–3 Hours

Interviews: 1 Interviews per Week on Completed Tools

◆ WEEK 1 (Days 1–6) — Linux Foundations

Mapped Sections:

Section 1–4

Topics

- Linux fundamentals
- File system hierarchy
- Users, groups, permissions
- Process & service management
- Logs & troubleshooting (journalctl, /var/log)
- Disk usage & monitoring

Projects (1)

Linux Server Monitoring & Audit Script

You will create a production-style monitoring script that:

- Checks CPU, memory, and disk usage
- Lists top 5 memory-consuming processes
- Checks failed login attempts
- Stores output in a timestamped log file

This project ensures you understand Linux troubleshooting and server health monitoring.

Interview Focus

- Linux troubleshooting
 - Permission issues
 - High CPU debugging
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◆ WEEK 2 (Days 7–12) — Git & Networking Basics

Mapped Sections:

Section 5, Section 9

Topics

- Git clone, commit, push, pull
- Branching & merging
- Merge conflicts & rebase
- OSI & TCP/IP basics
- DNS, CIDR
- ping, traceroute, netstat, ss

Projects (2)

Git Workflow Implementation

- Create main, dev, and feature branches
- Perform merge conflict resolution
- Practice rebase
- Create version tag (v1.0)

This simulates real company Git workflow.

Networking Troubleshooting Lab

- Test DNS resolution
- Identify open ports
- Simulate blocked port & debug
- Use networking commands to diagnose connectivity

Interview Focus

- Git branching strategy
 - Networking debugging scenarios
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◆ WEEK 3 (Days 13–18) — Bash, YAML & VM Lab

Mapped Sections:

Section 6, 7, 10

Topics

- Bash scripting (loops, conditions, functions)
- Cron jobs
- YAML structure
- VM setup (Vagrant / EC2 Linux)

Projects (1)

Automated Backup & Log Cleanup System

You will:

- Write a bash script to take directory backup
- Compress backup file
- Delete backups older than 7 days
- Schedule cron job
- Use YAML file for configuration input

This builds automation skills required in production.

Interview Focus

- Cron failures
 - Automation scripting scenarios
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◆ WEEK 4 (Days 19–24) — AWS Core Services

Mapped Sections:

Section 13

Topics

- AWS Global Infrastructure
- IAM users, roles, policies

- EC2, EBS, AMI
- S3 storage classes
- VPC (public/private subnets)
- Security Groups

Projects (1)

Deploy Web Application in Custom VPC

You will:

- Create custom VPC
- Create public & private subnets
- Launch EC2 in public subnet
- Attach IAM role
- Configure security group
- Install Nginx & host sample webpage
- Store static content in S3

This builds AWS networking & security foundation.

Interview Focus

- IAM best practices
 - VPC explanation
 - EC2 troubleshooting
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◆ WEEK 5 (Days 25–30) — AWS Advanced Architecture

Mapped Sections:

Section 14, 15

Topics

- ALB & Auto Scaling
- Route53
- RDS basics
- High Availability concepts
- Lift & Shift vs Re-architecture

Projects (1)

Highly Available AWS Architecture

You will:

- Deploy 2 EC2 instances
- Configure ALB
- Enable Auto Scaling
- Connect to RDS
- Test failover scenario

This prepares you for architecture-level interviews.

Interview Focus

- HA & DR
- Auto Scaling explanation

(Interviews start — 2 per week)

◆ WEEK 6 (Days 31–36) — CI/CD (Jenkins & GitHub Actions)

Mapped Sections:

Section 16–18

Topics

- CI/CD concepts
- Jenkins architecture
- Jenkinsfile
- Credentials & plugins
- GitHub Actions workflows

Projects (2)

Jenkins CI Pipeline

- Connect Git repo
- Build application
- Run test
- Archive artifacts

GitHub Actions Workflow

- Trigger on push
- Build Docker image
- Push to Docker Hub

Interview Focus

- Pipeline failures
 - Jenkins vs GitHub Actions
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◆ WEEK 7 (Days 37–42) — Terraform

Mapped Sections:

Section 21

Topics

- Providers
- Variables
- Outputs
- State file
- Modules

Projects (1)

Provision AWS Infrastructure Using Terraform

You will:

- Create VPC
- Create EC2
- Create Security Group
- Use variables
- Store state in S3 backend

Interview Focus

- Terraform state management
 - Drift detection
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◆ WEEK 8 (Days 43–48) — Ansible

Mapped Sections:

Section 22

Topics

- Inventory
- Playbooks
- Roles
- Handlers

Projects (1)

Server Configuration Using Ansible

You will:

- Install Nginx
- Deploy application
- Restart service via handler
- Maintain idempotency

Interview Focus

- Ansible vs Terraform
 - Configuration management
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◆ WEEK 9 (Days 49–54) — Docker

Mapped Sections:

Section 27

Topics

- Docker architecture
- Dockerfile best practices
- Multi-stage builds
- Docker Compose

Projects (1)

Dockerize Application

You will:

- Write optimized Dockerfile
- Use multi-stage build
- Run container
- Use Docker Compose for multi-service setup

Interview Focus

- Docker vs VM
 - Image optimization
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◆ **WEEK 10 (Days 55–60) — Kubernetes Core**

Mapped Sections:

Section 28

Topics

- K8s architecture
- Pods
- Deployments
- Services
- ConfigMaps & Secrets

Projects (1)

Deploy Application on Kubernetes

You will:

- Create deployment YAML
- Expose using Service
- Inject environment variables
- Perform rolling update

Interview Focus

- Kubernetes architecture explanation
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◆ WEEK 11 (Days 61–75) — Kubernetes Advanced

Mapped Sections:

Section 29, 30

Topics

- Ingress
- HPA
- Resource limits
- CrashLoopBackOff debugging

Projects (1)

Production-Ready Kubernetes Deployment

You will:

- Configure Ingress
- Set resource requests & limits
- Configure HPA
- Simulate pod failure & debug

Interview Focus

- Production K8s troubleshooting
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◆ WEEK 12 (Days 76–90) — Monitoring & Final Capstone

Mapped Sections:

Section 23, 24, 31, 32

Topics

- Prometheus
- Node exporter
- Alertmanager
- Grafana dashboards
- Full revision

- Resume walkthrough
- Scenario-based questions

Final Project (1)

End-to-End DevOps Pipeline

Flow:

Git → Jenkins → Docker → Kubernetes → Prometheus → Grafana

You must implement:

- CI/CD automation
- Containerized deployment
- Monitoring dashboard
- Alert on high CPU usage
- Rollback strategy

Interview Focus

- End-to-end explanation
- Real production outage scenarios